```
# Joint values (K_ij, H_ij) per inner element, cycle 0, exact integers. K =
<r_i,r_j>, H = <r_i,s_j>, s_j = partner of r_j.
import numpy as np, itertools, json
rt=[]
for i,j in itertools.combinations(range(8),2):
    for si in (2,-2):
        for sj in (2,-2):
            v=np.zeros(8,int); v[i]=si; v[j]=sj; rt.append(v)
for s in itertools.product([1,-1],repeat=8):
    if sum(x<0 for x in s)%2==0: rt.append(np.array(s))
G=(np.array(rt)@np.array(rt).T)//4; del rt                      # same table, same
ordering as scripts 45-51
part={int(k):v for k,v in json.load(open('51_cycle0_pairing.json')).items()}
r=np.array(sorted(part)); s=np.array([part[i] for i in r])
K=G[np.ix_(r,r)]; H=G[np.ix_(r,s)]; S2=G[np.ix_(s,s)]
print("H symmetric:",(H==H.T).all()," | <s_i,s_j> = K+H exactly:",(S2==K+H).all())
J=np.arange(240); J[r]=s; J[s]=r
D=G[np.ix_(J,J)]-G
print("Delta inner-inner = H:",(D[np.ix_(r,r)]==H).all()," | outer-outer = -H:",
(D[np.ix_(s,s)]==-H).all()," | cross blocks zero:",(D[np.ix_(r,s)]==0).all())
print("H diagonal (r_i with its own partner):",sorted(set(np.diag(H))))
joint={}
for a,b in zip(K.flatten(),H.flatten()):
joint[(int(a),int(b))]=joint.get((int(a),int(b)),0)+1
print("\njoint (K,H) totals over all 120x120:",dict(sorted(joint.items())))
pairsKH=sorted(joint)
rows=np.array([[int(((K[i]==a)&(H[i]==b)).sum()) for a,b in pairsKH] for i in
range(120)])
print("\nper-row counts (min..max) for each (K,H):")
for c,(a,b) in enumerate(pairsKH): print(f"   (K,H)=({a:2d},{b:2d}): per row
{rows[:,c].min()}..{rows[:,c].max()}  total {rows[:,c].sum()}")
print("\nevery row has the identical profile:",(rows==rows[0]).all())
```